hCRI.io API

Reference for the personal-token API used by scripts, instruments and the hCRI Companion app (app 1.25). Last updated October 10, 2026.

Overview

The API lets you upload spectra, attach flicker readings, get a share link, and read or set the LEDs (brand, LED, CCT for each) and light details of a report (optic, lumens, current, notes), all with a personal API token. Every endpoint returns JSON. Requests with a body use JSON unless noted (the upload endpoint also accepts a file).

Base URL `https://www.hcri.io/index.php/api/v1`

Authentication Create a token in *Profile* → *API* and send it in every request:

```
Authorization: Bearer hcri_YOUR_TOKEN
```

Tokens never expire and can be revoked at any time on the same page. A token carries your account's upload access, so treat it like a password. A token can only act on reports you own, and each endpoint below is deliberately narrow: an upload token cannot edit or delete reports, change account settings, or read other people's data.

Endpoint summary

Method	Path	Auth	Purpose
POST	/upload	Token	Upload a spectrum file; creates a report (or updates one of yours)
GET	/led_lists	None	LED brand / model / CCT lists (cacheable)
POST	/led_suggest	Token	Suggest the LED from a spectrum (and optional title)
POST	/reports/{id}/led	Token	Set the report's LEDs (brand, LED, CCT), light, optic, lumens, current, notes
POST	/reports/{id}/share	Token	Get a link to your report
POST	/flicker	Token	Create a flicker reading
GET	/flicker, /flicker/{id}	Token	List or read flicker readings
PATCH	/flicker/{id}	Token	Change label, notes, or attached report
DELETE	/flicker/{id}	Token	Delete a flicker reading

Upload a report

`POST` `/upload`

Send a measurement file as `multipart/form-data` field `file`, or POST the raw body with the `X-Filename` header (or `?filename=`) so the format can be detected. Accepted formats: CSV, TSV, TXT, JSON, SP and PDF, up to 10 MB. Spectra should cover about 380–780 nm. New uploads follow your Profile → Privacy default for Explore visibility.

Field	Where	Description
file	multipart	The measurement file (required unless the raw body is used)
label	form, query, or X-Label header	Report title (optional; defaults to the file name)
replaceId	form, query, or X-Replace-Id header	Optional id of one of your own reports to overwrite instead of creating a new one. Keeps its id, link, visibility, notes and LEDs; replaces the title and spectrum. Returns 200 with "replaced": true (an unknown or someone else's id creates a new report, "replaced": false).

```
curl -X POST "https://www.hcri.io/index.php/api/v1/upload" \
  -H "Authorization: Bearer YOUR_TOKEN" \
  -F "file=@measurement.csv" \
  -F "label=My Light"
```

Response 201 (200 when replaceId updated a report)

```
{ "id": 1234, "label": "My Light", "sourceType": "csv",
  "cct": 4012, "duv": 0.0021, "x": 0.3811, "y": 0.3794,
  "Rf": 92, "Rg": 99, "ra": 94.8, "r9": 71.2,
  "isPublic": false, "createdAt": "2026-10-10 15:50:38", "viaApi": true,
  "replaced": false }
```

Errors: 400 no file, 401 bad token, 405 wrong method, 413 file over 10 MB, 422 file could not be parsed.

LED brand, model and CCT

These endpoints power the LED suggestion in the app and the Fill in dialog on the site. A suggestion compares the spectrum with reports that already have an LED brand and model, and with the words in the reading's title. Only brand, model and CCT strings and aggregate scores are returned: never a report id, title, owner or spectrum. A match that rests only on one other person's private reports is never returned.

GET `/led_lists`

Public and read-only. Returns the curated dropdown values, plus `modelsByBrand`: suggested pairings of which LED models have been seen with which brand, built from how people have tagged their reports (most-used first). They are suggestions, not a fixed catalogue: a model that has not yet been tagged under a brand will not be listed for it. The response carries a `version` and an `ETag`; send the ETag back in `If-None-Match` and an unchanged list answers 304 with no body.

```
{ "lists": { "led_brand": ["Nichia", "Luminus", ...],
             "led_model": ["519A", "SST-20", ...],
             "led_cct":   ["2700K", "3000K", "4000K", ...] },
  "modelsByBrand": { "Nichia": ["519A", "219C"], ... },
  "version": "3f9a0c1d5b7e2a44" }
```

POST `/led_suggest`

Body: `wls` and `vals` (matching arrays, 10 to 5000 points) and an optional `title` (the reading's title, used as a hint).

```
curl -X POST "https://www.hcri.io/index.php/api/v1/led_suggest" \
  -H "Authorization: Bearer YOUR_TOKEN" -H "Content-Type: application/json" \
  -d '{"wls":[380,385,...],"vals":[0.01,0.02,...],"title":"Convoy S2+ 519A 4500K"}'
```

```
{ "suggestion": { "brand": "Nichia", "model": "519A", "cct": "4500K",
                  "score": 0.9712, "confidence": 0.74, "support": 5,
                  "source": "spectrum" },
  "alternatives": [ { "brand": "...", "model": "...", "cct": "...",
                      "score": 0.91, "confidence": 0.18, "support": 2 } ],
  "cctGuess": "4500K" }
```

Field	Meaning
suggestion	The clear winner, or null when nothing is clearly ahead
source	spectrum (curve match), title (a model named in the title that has tagged reports), or cct (nominal CCT only; brand and model empty)
score	Curve overlap, 0 to 1 (a suggestion needs 0.94 or more from the curve)
confidence	Share of the evidence behind this brand and model, 0 to 1
support	Number of very close tagged reports behind it
cctGuess	Nearest nominal CCT from the list for the measured CCT, or null

POST `/reports/{id}/led`

Sets the LED details on a report you uploaded. A light can have more than one LED, each with its own brand, LED (model) and CCT, so they stay paired. Send them as a list in `leds`: each entry has `brand`, `led` (the LED model; `model` is accepted too) and `cct`, any of which may be left out (a deep-red LED has no CCT). The list replaces the report's LEDs, and an empty list removes them all. For a single LED you can still send `brand`, `led`/`model` and `cct` directly; that sets the first LED and leaves any others alone.

```
curl -X POST "https://www.hcri.io/index.php/api/v1/reports/1234/led" \
  -H "Authorization: Bearer YOUR_TOKEN" -H "Content-Type: application/json" \
  -d '{"leds":[{"brand":"Nichia","led":"519A","cct":"1800K"},{"brand":"Luminus","led":"SST-20-DR"}]}'
```

The same body can also carry, as single values: `light_brand`, `light_model`, `optic`, `lumens`, `current`, and `notes` (the report's notes text, up to 10,000 characters; an empty string clears it). Values already in the curated lists are saved. New values are queued for admin review instead of being added to the lists directly, except `lumens` and `current`, which are free-form numbers and are saved as sent.

```
{ "leds":      [ { "brand": "Nichia",  "led": "519A",      "cct": "1800K" },
                 { "brand": "Luminus", "led": "SST-20-DR", "cct": null } ],
  "saved":     { "led_brand": "Nichia", "led_model": "519A", "led_cct": "1800K" },
  "requested": { } }
```

A value that is not in the lists is left blank in that LED and listed under requested (as an array per field when you send leds). A report's own GET response also carries its leds.

Errors: 400 invalid id, 401 bad token, 403 not your report, 404 report not found.

Share link

POST `/reports/{id}/share`

Returns a link to a report you own. A public report links to `https://www.hcri.io/?report={id}` and needs no token. A private report gets a share token, reused on later calls so a link you already handed out keeps working.

```
{ "link": "https://www.hcri.io/?share=9f3c1a7b2d4e5f60", "isPublic": false,
  "shareToken": "9f3c1a7b2d4e5f60" }
```

Errors: 400 missing id, 401 bad token, 404 not found (also returned for a report that belongs to someone else).

Flicker readings

A flicker reading holds the frequency, modulation depth, flicker index, cycle time and the captured waveform. A reading belongs to at most one report. Deleting a report keeps its readings and detaches them.

POST `/flicker`

Field	Type	Description
waveform	number[] (2–4000)	Required. Captured samples
frequencyHz	number	Flicker frequency
percentFlicker	number	Percent flicker (modulation depth)
flickerIndex	number	Flicker index
cycleMs	number	One cycle, in milliseconds
spanMs	number	Length of the whole capture, in milliseconds
label	string	Title (defaults to "Flicker YYYY-MM-DD HH:MM")
notes	string	Free text
model	string	Meter model

Field	Type	Description
capturedAt	ISO 8601	When it was captured
settings	object	Capture settings, stored as given
reportId	integer	Attach to one of your reports

```
{ "success": true, "id": 87, "reportId": 1234 }      // HTTP 201
```

GET `/flicker · /flicker/{id}`

The list returns your readings without waveforms (most recent first, up to 500); filter with `?unattached=1` or `?report_id=N`. A single reading includes its waveform. You can read your own readings, and readings attached to a report you are allowed to view.

PATCH `/flicker/{id}`

Change `label` (cannot be empty), `notes`, or `reportId` (null detaches). Returns the updated reading.

DELETE `/flicker/{id}`

Deletes one of your readings.

Errors: 400 invalid value (the message names the field), 401 bad token, 404 not found.

Errors and limits

Status	Meaning
400	Missing or invalid input; the JSON body {"error": "…"} says what
401	Missing or invalid token
403	The report is not yours
404	Not found (also used for other people's reports)
405	Method not allowed for this endpoint
413	Upload larger than 10 MB
422	The uploaded file could not be parsed

Errors are JSON: `{"error": "message"}`. Responses are not cached except `/led_lists`. There is no published rate limit; please keep scripts to a reasonable pace.

A browser tester for uploads is at `/api-test.html`.